

Medical milestones

In the 18th and 19th centuries, technological innovations and the growing scientific approach taken to medicine led to many important advances in the field. During this period, new inventions, treatments, and methods of diagnosis laid the foundation for modern medical care.

Some of the new knowledge was used to improve on current methods of treating disease. In parts of Asia, Africa, and the Middle East, a process called “variolation,” in which a mild form of a disease was introduced into a healthy body to immunize it, had been used for centuries against smallpox. However, this was unreliable. In 1796, the first safe and reliable vaccine for smallpox was created in Britain.

New discoveries about the natural world led to an improved understanding of human biology. Some

tiny microorganisms, such as varieties of bacteria, were found to be capable of causing disease. These organisms became known as germs. Antiseptics—substances that kill germs—were developed, and as a result nursing methods improved significantly.

In 1901, Austrian biologist Karl Landsteiner’s discovery of the four types of human blood groups A, B, O, and AB was crucial to improving the success of blood transfusions, and would change medical surgery forever.

MARY SEACOLE

When British-Jamaican nurse Mary Seacole (1805–1881) offered her nursing services during the Crimean War (1853–1856), she was rejected because of her skin color. However, she used her own money and traveled to Crimea to tend to the wounded British soldiers.

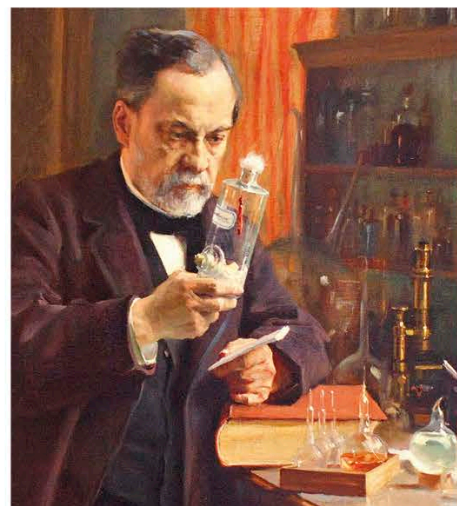


FACT

British nurse Florence Nightingale established the first-ever nursing school called “The Nightingale Training School for Nurses” at St. Thomas’ Hospital in London in 1860.

THE FIRST VACCINATION ►

In 1796, English doctor Edward Jenner observed that milkmaids who caught cowpox did not catch smallpox (a much more dangerous disease). Jenner injected a boy with cowpox—he fell slightly ill but recovered. Jenner then exposed the boy to smallpox—he suffered only a mild form of the disease. This vaccine eventually led to the complete elimination of the disease.



▲ GERM THEORY

French biochemist Louis Pasteur discovered that microorganisms in the air made wine turn sour. He then looked at how microorganisms affected the human body, and, in 1861, he published his findings that germs invading human bodies caused many diseases. German microbiologist Robert Koch later identified many specific bacteria that caused individual diseases.

